

CUSTOMER PERSONALITY ANALYSIS

Task 1: Data Collection and Data Understanding

Project Report

Dataset Source: Kaggle — Customer Personality Analysis
File: marketing_campaign.csv

Status: Completed
Priority: High

Objective

The objective of this activity is to download, inspect, understand, and document the Customer Personality Analysis dataset before performing any preprocessing or analysis. This report walks through the dataset's structure, quality, and business relevance, and establishes a documented foundation (data dictionary and data quality report) for subsequent cleaning and analysis phases.

Task 1 & 2 — Data Collection and Project Organization

1.1 Downloading the Dataset

The Customer Personality Analysis dataset was downloaded directly from Kaggle using the Kaggle API (dataset: imakash3011/customer-personality-analysis). The downloaded archive was extracted programmatically, and successful extraction was verified by listing the contents of the target folder, which confirmed the presence of the raw file `marketing_campaign.csv`.

1.2 Verification of Download

The dataset was loaded into a pandas DataFrame and inspected using `.shape`, `.head()`, and `.dtypes` to confirm that the file was read correctly and that the expected 2,240 records and 29 columns were present.

1.3 Project Folder Structure

A dedicated project folder was created to store all files related to this activity in an organized, auditable structure:

- 01_Raw_Data — the original, unmodified `marketing_campaign.csv` file
- 02_Inspection_Report — notes and outputs from dataset inspection (Tasks 3–9)
- 03_Data_Quality — findings on missing values, duplicates, and inconsistent values
- 04_Data_Dictionary — the variable-level data dictionary (Task 12)
- 05_Screenshots — screenshots evidencing the download and folder structure
- 06_Final_Report — this consolidated report and supporting summary documents

Deliverables for this stage: the raw dataset file, a screenshot of the successful download, and a screenshot of the folder structure described above.

Task 3 — Basic Dataset Inspection

The dataset was inspected at a high level using Python (pandas) to confirm its basic structural properties before any deeper analysis.

Property	Value
Rows	2,240
Columns	29
File Type	CSV (tab-separated values)
Dataset Size	215.03 KB (220,188 bytes)
Encoding	ASCII

Note: although the file has a .csv extension, its fields are separated by tab characters rather than commas, so it must be read with `sep="\t"` in pandas. This is itself a minor documentation gap worth flagging for downstream users of the file.

Task 4 — Variable Inspection

All 29 variables were reviewed to record their data type and business meaning. Of the 29 columns, 26 are numerical (integer or float) and 2 are stored as text/categorical (Education, Marital_Status); the remaining categorical/binary flags — such as the campaign-acceptance and Response fields — are numerically encoded as 0/1. One column (Dt_Customer) stores a date as text.

Variable	Data Type	Description (Understanding)
ID	Numerical (Integer)	Unique identifier assigned to each customer
Year_Birth	Numerical (Integer)	Customer's year of birth
Education	Categorical (Text)	Customer's highest level of education
Marital_Status	Categorical (Text)	Customer's marital / relationship status
Income	Numerical (Float)	Customer's yearly household income
Kidhome	Numerical (Integer)	Number of small children in the customer's household
Teenhome	Numerical (Integer)	Number of teenagers in the customer's household
Dt_Customer	Date	Date the customer enrolled with the company
Recency	Numerical (Integer)	Number of days since the customer's last purchase
MntWines	Numerical (Integer)	Amount spent on wine products in the last 2 years
MntFruits	Numerical (Integer)	Amount spent on fruit products in the last 2 years
MntMeatProducts	Numerical (Integer)	Amount spent on meat products in the last 2 years
MntFishProducts	Numerical (Integer)	Amount spent on fish products in the last 2 years
MntSweetProducts	Numerical (Integer)	Amount spent on sweet products in the last 2 years
MntGoldProds	Numerical (Integer)	Amount spent on gold / premium products in the last 2 years
NumDealsPurchases	Numerical (Integer)	Number of purchases made using a discount / deal
NumWebPurchases	Numerical (Integer)	Number of purchases made through the company website
NumCatalogPurchases	Numerical (Integer)	Number of purchases made using a catalogue
NumStorePurchases	Numerical (Integer)	Number of purchases made directly in physical stores
NumWebVisitsMonth	Numerical (Integer)	Number of visits to the company website in the last month
AcceptedCmp1	Categorical (Binary)	1 if customer accepted the offer in campaign 1, else 0
AcceptedCmp2	Categorical (Binary)	1 if customer accepted the offer in campaign 2, else 0
AcceptedCmp3	Categorical (Binary)	1 if customer accepted the offer in campaign 3, else 0
AcceptedCmp4	Categorical (Binary)	1 if customer accepted the offer in campaign 4, else 0

Variable	Data Type	Description (Understanding)
AcceptedCmp5	Categorical (Binary)	1 if customer accepted the offer in campaign 5, else 0
Complain	Categorical (Binary)	1 if the customer complained in the last 2 years, else 0
Z_CostContact	Numerical (Integer, constant)	Fixed cost of contacting a customer (constant value = 3)
Z_Revenue	Numerical (Integer, constant)	Fixed revenue from a successful campaign response (constant = 11)
Response	Categorical (Binary)	1 if customer accepted the offer in the most recent campaign, else 0

Task 5 — Identify Missing Values

Every column was checked for missing (null) values.

Variable	Missing Values	Percentage (%)
Income	24	1.07%

Columns without missing values: all remaining 28 columns — ID, Year_Birth, Education, Marital_Status, Kidhome, Teenhome, Dt_Customer, Recency, MntWines, MntFruits, MntMeatProducts, MntFishProducts, MntSweetProducts, MntGoldProds, NumDealsPurchases, NumWebPurchases, NumCatalogPurchases, NumStorePurchases, NumWebVisitsMonth, AcceptedCmp1–5, Complain, Z_CostContact, Z_Revenue, and Response — contain no missing values (0%).

Only Income has missing data, and at 1.07% (24 of 2,240 records) the impact is minor. It can reasonably be addressed later through median/mean imputation or row exclusion without materially affecting the dataset.

Task 6 — Identify Duplicate Records

The dataset was checked for fully duplicated rows and for duplicate Customer IDs.

- Duplicate rows (all columns identical): 0
- Duplicate Customer IDs: 0

Finding: No duplicate records exist in the dataset. Every customer ID is unique, and no two rows are identical across all 29 fields. This is a positive quality signal — no de-duplication step will be required in preprocessing.

Task 7 — Identify Incorrect or Inconsistent Values

7.1 Invalid Ages / Birth Years

Filtering for implausible Year_Birth values revealed 3 records with birth years far outside a realistic range: 1893, 1899, and 1900. These would correspond to customer ages of 120+ years as of the campaign period (2012–2014), which is not plausible and indicates data-entry error or placeholder values.

ID	Year_Birth	Implied Age (~2014)	Education	Marital_Status
11004	1893	121	2n Cycle	Single
1150	1899	115	PhD	Together
7829	1900	114	2n Cycle	Divorced

7.2 Negative Income

The Income column was checked for negative values. No negative income values were found; the minimum recorded income is 1,730, which is a valid (if low) positive value.

7.3 Empty / Blank Categories

Education and Marital_Status were checked for blank or whitespace-only entries. No empty category values were found in either column.

7.4 Incorrect Dates

Dt_Customer was parsed using the expected day-month-year format (%d-%m-%Y). All 2,240 values parsed successfully with no invalid or unparseable dates. The resulting enrollment dates range from 30 July 2012 to 29 June 2014, which is an internally consistent window.

7.5 Wrong Spellings / Unexpected Values

Education contains 5 clean, consistently spelled categories (Graduation, PhD, Master, 2n Cycle, Basic) with no spelling issues. Marital_Status, however, contains three implausible or non-standard categories in addition to the five expected ones:

- "Alone" (3 records) — likely a duplicate/synonym of "Single"
- "Absurd" (2 records) — not a valid marital status; likely a placeholder or joke entry
- "YOLO" (2 records) — not a valid marital status; likely a placeholder or joke entry

These 7 records (0.31% of the dataset) should be corrected or consolidated during data cleaning — for example, mapping "Alone" to "Single" and treating "Absurd"/"YOLO" as missing or as "Single"/"Other" depending on business judgement.

7.6 Summary of Issues Found

- 3 records with implausible birth years (customers appearing 114–121 years old)
- 7 records with non-standard Marital_Status values ("Alone", "Absurd", "YOLO")

- 1 extreme income outlier (666,666 — see Task 8) that is far outside the normal range and warrants verification
- 24 missing Income values (see Task 5)
- File is tab-delimited despite the .csv extension, which can cause read errors if not handled explicitly

Task 8 — Review of Numerical Variables

The dataset contains 26 numerical columns (24 shown below plus the two constant columns `Z_CostContact` and `Z_Revenue`, which take a single fixed value across all records — 3 and 11 respectively — and add no analytical variance). Minimum, maximum, and mean values were computed for each.

Variable	Minimum	Maximum	Mean	Observation
Year_Birth	1893	1996	1968.81	Contains 3 implausible birth years (1893, 1899, 1900)
Income	1,730	666,666	52,247.25	Extreme outlier at 666,666; right-skewed distribution
Kidhome	0	2	0.44	Small integer range; no anomalies
Teenhome	0	2	0.51	Small integer range; no anomalies
Recency	0	99	49.11	Evenly distributed; no anomalies
MntWines	0	1,493	303.94	Right-skewed; wine is the highest-spend category
MntFruits	0	199	26.30	Right-skewed; low-value category
MntMeatProducts	0	1,725	166.95	Right-skewed; second-largest spend category
MntFishProducts	0	259	37.53	Right-skewed; low-value category
MntSweetProducts	0	263	27.06	Right-skewed; low-value category
MntGoldProds	0	362	44.02	Right-skewed; moderate-value category
NumDealsPurchases	0	15	2.33	No anomalies
NumWebPurchases	0	27	4.08	No anomalies
NumCatalogPurchases	0	28	2.66	No anomalies
NumStorePurchases	0	13	5.79	Highest-volume purchase channel on average
NumWebVisitsMonth	0	20	5.32	No anomalies

Key observations: Income is strongly right-skewed with a single extreme outlier at 666,666 (roughly 10x the next-highest value of 162,397) that should be investigated and likely treated as an error or capped before modelling. Spending columns (MntWines through MntGoldProds) are all right-skewed, which is expected for retail spend data — most customers spend modestly, while a smaller group of high-value customers drives the upper range. Wine and meat products account for the largest share of spending on average.

Task 9 — Review of Categorical Variables

Education and Marital_Status are the two variables stored as free text; several additional columns are categorical in nature but numerically encoded as binary (0/1) flags.

Variable	Number of Categories	Categories (with counts)
Education	5	Graduation (1,127), PhD (486), Master (370), 2n Cycle (203), Basic (54)
Marital_Status	8	Married (864), Together (580), Single (480), Divorced (232), Widow (77), Alone (3), Absurd (2), YOLO (2)
AcceptedCmp1 / 2 / 3 / 4 / 5 (binary)	2 each	0 = did not accept, 1 = accepted
Complain (binary)	2	0 = no complaint, 1 = complaint
Response (binary)	2	0 = did not respond, 1 = responded to last campaign

Task 10 — Business Understanding

Based on the structure and content of the dataset, the following business indicators were identified as important for downstream analysis, segmentation, and marketing decision-making:

High-Income Customers

Customers with Income above roughly the 75th percentile (~68,500). These customers have the greatest spending potential and are prime targets for premium product lines and higher-value offers.

Frequent Buyers

Customers with a high combined purchase count across web, catalogue, and store channels (NumWebPurchases + NumCatalogPurchases + NumStorePurchases). They represent the most engaged, revenue-generating segment.

Campaign Responders

Customers who accepted one or more of the five historical campaigns (AcceptedCmp1–5) or the most recent campaign (Response = 1). This group indicates which customers are receptive to promotional offers.

Customers with Complaints

Customers where Complain = 1. This is a small but important group for service-recovery and churn-prevention efforts, since dissatisfaction can precede customer loss.

Website Visitors

Customers with high NumWebVisitsMonth. Frequent site visits but low NumWebPurchases may indicate browsing-without-converting behaviour worth investigating.

High-Spending Customers

Customers with a high total spend across all six product categories (Mnt*). This is a key indicator of overall customer value (CLV).

Recently Active Customers

Customers with low Recency (few days since last purchase). Low recency signals an actively engaged customer, while high recency may signal churn risk.

Task 11 — Initial Data Summary

Dataset Overview

The Customer Personality Analysis dataset contains records of a retail company's customers, covering demographic attributes, product spending, purchase channels, campaign responses, and complaints. It is intended to support customer segmentation and targeted marketing analysis.

Metric	Value
Number of Variables	29
Number of Records	2,240
Numerical Variables	26
Categorical (Text) Variables	2 (Education, Marital_Status)
Date Variables	1 (Dt_Customer)
Binary Flag Variables	8 (AcceptedCmp1–5, Complain, Response, and encoded within numerical count)
Missing Values	24 (Income only, 1.07%)
Duplicate Records	0

Major Data Quality Issues

- Income: 24 missing values (1.07%) and 1 extreme outlier (666,666)
- Year_Birth: 3 implausible birth years (1893, 1899, 1900)
- Marital_Status: 7 records with non-standard values ("Alone", "Absurd", "YOLO")
- File format: tab-delimited despite the .csv extension

Overall, the dataset is of good quality: it is free of duplicate records, has minimal missingness confined to a single column, and its date and category fields are largely clean. The issues identified are limited in scope and can be resolved with standard cleaning techniques (imputation, outlier treatment, and category standardization) before the dataset is used for segmentation or predictive modelling.

Task 12 — Data Dictionary

The data dictionary below documents every variable in the dataset, its data type, and its business meaning, to support consistent understanding across the project team.

Variable	Description	Data Type	Business Meaning
ID	Unique identifier assigned to each customer	Numerical (Integer)	Primary key used to identify and track individual customers
Year_Birth	Customer's year of birth	Numerical (Integer)	Used to derive customer age for demographic segmentation
Education	Customer's highest level of education	Categorical (Text)	Indicates socio-educational profile; useful for targeted messaging
Marital_Status	Customer's marital / relationship status	Categorical (Text)	Reflects household and family situation; supports segmentation
Income	Customer's yearly household income	Numerical (Float)	Key indicator of purchasing power and spending potential
Kidhome	Number of small children in the customer's household	Numerical (Integer)	Household composition indicator; affects product relevance
Teenhome	Number of teenagers in the customer's household	Numerical (Integer)	Household composition indicator; affects product relevance
Dt_Customer	Date the customer enrolled with the company	Date	Used to compute customer tenure and loyalty
Recency	Number of days since the customer's last purchase	Numerical (Integer)	Measures recent engagement; low values indicate active customers
MntWines	Amount spent on wine products in the last 2 years	Numerical (Integer)	Product-category spending; identifies wine-focused customers
MntFruits	Amount spent on fruit products in the last 2 years	Numerical (Integer)	Product-category spending indicator
MntMeatProducts	Amount spent on meat products in the last 2 years	Numerical (Integer)	Product-category spending; typically the 2nd largest category
MntFishProducts	Amount spent on fish products in the last 2 years	Numerical (Integer)	Product-category spending indicator
MntSweetProducts	Amount spent on sweet products in the last 2 years	Numerical (Integer)	Product-category spending indicator

Variable	Description	Data Type	Business Meaning
MntGoldProds	Amount spent on gold / premium products in the last 2 years	Numerical (Integer)	Indicates affinity for premium or gift-type products
NumDealsPurchases	Number of purchases made using a discount / deal	Numerical (Integer)	Identifies deal-sensitive, price-conscious customers
NumWebPurchases	Number of purchases made through the company website	Numerical (Integer)	Measures engagement with the online sales channel
NumCatalogPurchases	Number of purchases made using a catalogue	Numerical (Integer)	Measures engagement with the catalogue sales channel
NumStorePurchases	Number of purchases made directly in physical stores	Numerical (Integer)	Measures engagement with the in-store sales channel
NumWebVisitsMonth	Number of visits to the company website in the last month	Numerical (Integer)	Proxy for online browsing interest / engagement
AcceptedCmp1	1 if customer accepted the offer in campaign 1, else 0	Categorical (Binary)	Tracks responsiveness to a specific historical marketing campaign
AcceptedCmp2	1 if customer accepted the offer in campaign 2, else 0	Categorical (Binary)	Tracks responsiveness to a specific historical marketing campaign
AcceptedCmp3	1 if customer accepted the offer in campaign 3, else 0	Categorical (Binary)	Tracks responsiveness to a specific historical marketing campaign
AcceptedCmp4	1 if customer accepted the offer in campaign 4, else 0	Categorical (Binary)	Tracks responsiveness to a specific historical marketing campaign
AcceptedCmp5	1 if customer accepted the offer in campaign 5, else 0	Categorical (Binary)	Tracks responsiveness to a specific historical marketing campaign
Complain	1 if the customer complained in the last 2 years, else 0	Categorical (Binary)	Flags dissatisfied customers for service-recovery efforts
Z_CostContact	Fixed cost of contacting a customer (constant value = 3)	Numerical (Integer, constant)	Used in campaign profitability / ROI calculations
Z_Revenue	Fixed revenue from a successful campaign response (constant = 11)	Numerical (Integer, constant)	Used in campaign profitability / ROI calculations
Response	1 if customer accepted the offer in the most recent campaign, else 0	Categorical (Binary)	Target variable for predicting campaign response

Task 13 — Data Quality Report

The table below consolidates every data quality issue identified during inspection, together with a recommended resolution for the upcoming data preparation phase.

Issue	Variable	Description	Recommendation
Missing Values	Income	24 records (1.07%) have no recorded income value.	Impute using median income (robust to the outlier) or a segment-based median; alternatively exclude the 24 rows if modelling is sensitive to imputation bias.
Extreme Outlier	Income	One customer recorded with income of 666,666 — roughly 4x the next-highest value and 12x the mean.	Verify against source system if possible; otherwise cap (winsorize) or remove as an outlier before analysis/modelling.
Invalid / Implausible Values	Year_Birth	3 customers have birth years of 1893, 1899, and 1900, implying ages over 110 years.	Treat as data-entry errors; either exclude these rows or replace with a null/median value flagged for review.
Inconsistent Categories	Marital_Status	7 records contain non-standard values "Alone" (3), "Absurd" (2), and "YOLO" (2) alongside the 5 valid categories.	Standardize "Alone" into "Single"; recode "Absurd"/"YOLO" as "Single" or "Other" based on business input, or treat as missing.
File Format Ambiguity	N/A (whole file)	File has a .csv extension but is actually tab-delimited, which can cause silent read errors with default CSV parsers.	Document the delimiter explicitly (sep="\t") in all data-loading code and project documentation.
Low-Variance Columns	Z_CostContact, Z_Revenue	Both columns hold a single constant value (3 and 11 respectively) across all 2,240 records.	Retain for campaign ROI calculations but exclude from statistical/ML feature sets, as they carry no predictive variance.